RouteRec: Behavior-Guided Expert Routing for Sequential Recommendation

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Abstract

Sequential recommendation is traditionally modeled as a largely shared computation pipeline, failing to account for the inherent heterogeneity in user behavior. While Mixture-of-Experts (MoE) architectures have been introduced to enable adaptive computation, existing approaches predominantly rely on *hidden-state-driven routing*. This implicit routing mechanism entangles representation learning with routing decisions, making the basis of expert specialization opaque, hard to diagnose, and difficult to control.

We propose **RouteRec**, a behavior-guided MoE framework that explicitly decouples *why* an example is routed from *how* its representation is transformed. Unlike conventional MoE models, RouteRec derives compact, interpretable behavioral cues covering temporal pace, category focus, repetition memory, and popularity exposure from raw interaction logs and uses them exclusively to govern routing. By separating the routing signal from the sequence hidden state, RouteRec ensures that expert specialization is grounded in identifiable user-session characteristics. Furthermore, we organize routing across multiple temporal scales (macro, mid, and micro) to capture both persistent habits and transient intents. Extensive experiments demonstrate that RouteRec significantly outperforms state-of-the-art baselines, offering a more interpretable and controllable path for adaptive sequential recommendation.

CCS Concepts

• **Information systems** → **Recommender systems**.

Keywords

sequential recommendation, mixture of experts, conditional computation, routing, behavioral features

1 Introduction

Sequential recommendation is commonly formulated as predicting the next item from an ordered interaction history. While the prediction objective is straightforward, the behavioral structures underlying these sequences are often highly complex and non-stationary. A user may exhibit stable long-term preferences in one session, yet switch to rapid exploratory browsing or repetitive consumption in another. Despite this evident heterogeneity, most modern sequential recommenders [5, 7, 10] still rely on a "one-size-fits-all" shared computation pipeline, applying the same transformation logic to behaviorally distinct scenarios.

To address this, Mixture-of-Experts (MoE) has emerged as a promising paradigm for conditional computation [4, 16]. By allocating different inputs to specialized expert networks, MoE can theoretically adapt its capacity to diverse user behaviors. However, a critical limitation in current expert-based recommenders is their reliance on *implicit routing*. In these architectures, the routing gate is typically a function of the same hidden states used for representation. Because these hidden states are high-dimensional latent vectors where item semantics, position, and noise are heavily entangled, the resulting routing decisions often lack clear behavioral grounding. This makes it nearly impossible to determine whether an expert is specializing in a user's category-specific loyalty or merely reacting to a latent geometric artifact of the embedding space.

In this paper, we advocate for a different design principle: **Representation-Routing Decoupling**. Our core premise is that the signals determining *which* computation path to take (routing) should be distinct from the signals used to *transform* the content (representation). We observe that many critical behavioral regimes in recommendation are not purely latent; they can be captured through compact, explicit cues derived from ordinary interaction logs, such as interaction intervals, category switching rates, and repetition counts. By delegating routing to these explicit behavioral cues while reserving hidden states for expert transformation, we can make the specialization process more transparent and behavior-aware by construction.

Based on this idea, we propose **RouteRec**, a behavior-guided MoE framework for sequential recommendation. RouteRec constructs a bank of compact behavioral features—Tempo, Focus, Memory, and Exposure—to serve as the primary control signals for expert selection. To reflect that behavioral variation occurs at different temporal granularities, RouteRec organizes its routing across three stages: *Macro* routing for cross-session habits, *Mid* routing for within-session intent, and *Micro* routing for immediate transitions. This multi-stage design allows the model to synchronize its routing structure with the temporal scope of the evidence, transitioning from hierarchical group-based specialization to direct, fine-grained adaptation.

The main contributions of this work are as follows:

- We identify the pitfalls of implicit, hidden-state-driven routing in sequential MoE models and propose the principle of Representation-Routing Decoupling.
- We introduce RouteRec, which uses explicit behavioral cues as exclusive routing signals across multiple temporal scales (Macro-Mid-Micro).
- We provide an interpretable behavioral basis for expert specialization, demonstrating that lightweight cues from interaction logs can yield more effective routing than complex latent states.
- We show empirically that RouteRec consistently improves recommendation performance over strong Transformer, Mamba, and MoE-based baselines.

2 Related Work

2.1 Modeling Heterogeneous User States

Recognizing that user behavior is non-stationary, prior research has explored multi-interest extraction [2, 8] and short-term intent modeling [9, 12]. While these methods acknowledge behavioral heterogeneity, they typically address it at the *representation level* by designing more complex encoders or attention mechanisms to fuse diverse intents into a refined latent vector. In contrast, RouteRec addresses heterogeneity from the perspective of *computation-path separation*. Rather than trying to represent all behaviors within a single shared transformation, we physically route different behaviors to different experts, allowing for more explicit specialization.

2.2 Feature-Aware Sequential Recommendation

Many models incorporate auxiliary signals like timestamps, categories, or item side-information to strengthen sequence modeling [3, 10, 22, 23]. In these works, behavioral features are primarily treated as *representation enrichers*—they are added to input embeddings or used to bias self-attention within a shared pipeline. RouteRec departs from this by using behavioral features as *exclusive routing signals*. We do not use these cues to enrich the hidden state itself; instead, we use them as the "switch" that determines which specialized expert should process the standard sequence hidden state, preserving the clarity of each expert's role.

2.3 MoE and Conditional Computation

MoE has been applied to various recommendation tasks to scale model capacity efficiently [1, 11, 14, 19]. However, most existing MoE recommenders use hidden states or task-specific latent representations to drive the gating network. As discussed, this implicit routing leads to entangled specialization that is difficult to interpret or control. RouteRec is unique in its routing semantics: it shifts the routing control from implicit hidden states to explicit behavioral cues. This ensures that expert selection is synchronized with identifiable behavioral regimes (e.g., exploratory vs. repeat-heavy), making the model's adaptive computation behavior-aware by construction.

3 Methodology

3.1 Problem Setting and RouteRec Overview

Sequential recommendation predicts the next item from an ordered interaction prefix. We organize each user history into sessions and omit the user index for readability. Let $\mathcal{S} = [s_1, \ldots, s_M]$ denote the session sequence, and let $s_m = [x_1, \ldots, x_{|s_m|}]$ denote the current session. Each interaction x_t contains at least item identity and timestamp, and may include a coarse category label. Given observations up to position t, the model estimates the next-item distribution.

Figure 1 summarizes the overall RouteRec architecture and stage-wise routing design.

RouteRec keeps the SASRec backbone unchanged at the representation level [7], but replaces shared feed-forward transformation with stage-wise MoE routing [4, 6, 16]. The architectural principle is explicit role separation: stage features determine routing preference, while experts transform hidden representations. This separation allows the model to keep a strong common encoder while still allocating computational capacity according to behavioral context.

Let $\mathcal{T} = \{\text{macro}, \text{mid}, \text{micro}\}$ be the stage set, $\mathbf{h}_t$ be the hidden state at position t, and $\mathbf{f}_t^{(s)}$ be the stage feature used by router s. Route selection and expert aggregation are written as

$$\boldsymbol{\pi}_t^{(s)} = \text{softmax}\Big(\text{Router}_s(\mathbf{f}_t^{(s)})/\tau_s\Big), \tag{1}$$

$$\tilde{\mathbf{h}}_t^{(s)} = \sum_{e=1}^{E} \pi_{t,e}^{(s)} \, \text{Expert}_e^{(s)}(\mathbf{h}_t). \tag{2}$$

Equation (2) is shared across stages, but temporal meaning and routing structure differ. Macro routing captures broad cross-session tendency, mid routing adapts to evolving within-session context, and micro routing reacts to immediate transition signals. In this way, RouteRec aligns routing granularity with temporal scope instead of relying on a single fixed specialization rule.

3.2 Behavioral Features from Raw Signals

RouteRec is designed for realistic data availability. It uses signals that are commonly present in interaction logs: item identity, timestamp, optional category, and optional corpus-level popularity. We intentionally avoid heavy side-information encoders in the routing branch. Instead, raw signals are converted into compact scalar summaries that are easy to compute, stable across datasets, and interpretable at analysis time.

For session s_m and target position t, the stage windows are

$$\mathcal{R}_t^{\text{macro}} = [s_{\max(1,m-H)}, \ldots, s_{m-1}], \tag{3}$$

$$\mathcal{R}_t^{\text{mid}} = [x_1, \ldots, x_{t-1}], \tag{4}$$

$$\mathcal{R}_t^{\text{micro}} = [x_{\max(1,t-w)}, \ldots, x_{t-1}], \tag{5}$$

where $H = 5$ and $w = 5$ in our main setting. A shared summary operator $\mathcal{A}(\cdot)$ maps each window to a low-dimensional feature vector:

$$\mathbf{f}_t^{(s)} = \mathcal{A}(\mathcal{R}_t^{(s)}), \qquad s \in \mathcal{T}.$$

The summary dimensions are grouped into four semantic families: *Tempo*, *Focus*, *Memory*, and *Exposure*. Tempo captures pace and interval statistics. Focus captures concentration and switching behavior over coarse categories or item groups. Memory captures recurrence, repetition, and overlap with recent context. Exposure captures interactions with popularity strata and their local drift. This decomposition is semantic rather than architectural: it gives routing modules an interpretable behavioral basis.

Importantly, the feature template is shared across stages, but each stage reads it through a different temporal lens.

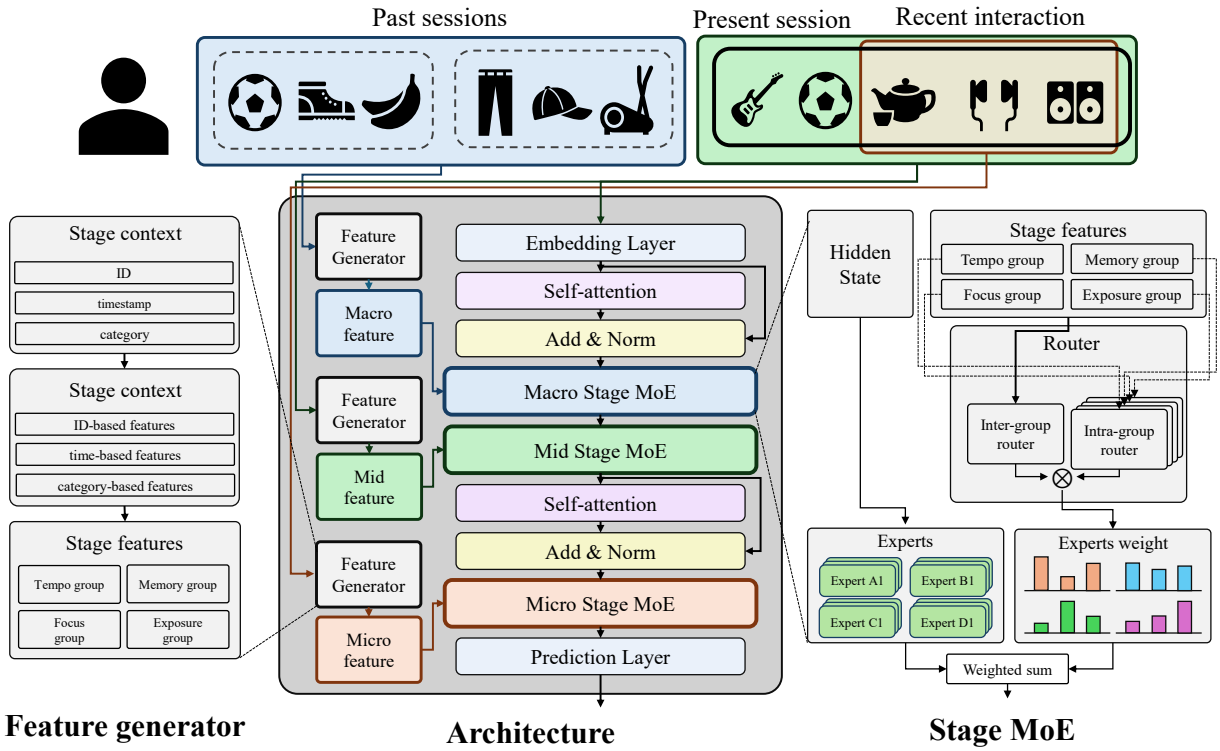

Figure 1: Overall RouteRec framework. (a) RouteRec keeps the SASRec backbone and replaces shared feed-forward computation with stage-wise mixture-of-experts routing; stage features determine routing weights, while expert FFNs transform hidden states. (b) The routing computation is built from three reusable components: group-level routing, conditional within-group expert routing, and direct expert routing. (c) The three stages combine these components differently: the macro stage uses hierarchical routing, the mid stage blends hierarchical and direct routing, and the micro stage uses direct expert routing.

Macro features reflect persistent tendency and broad preference shape. Mid features track evolving intent inside the current session. Micro features emphasize immediate cues around the prediction target. This shared-template, stage-specific-view design keeps the feature branch compact while preserving temporal specificity.

3.3 Feature-Guided Multi-Stage Routing

The same routing interface is used at all stages, but each stage adopts a different routing bias to match its temporal evidence quality.

Macro stage (group-first). Macro features are comparatively stable within one session and are better suited for coarse specialization than fine-grained instant decisions. We therefore use hierarchical group-first routing. Let $g(e)$ be the group index of expert e. Group logits and within-group logits are added in logit space:

$$q_{t,e}^{\text{macro}} = q_{t,g(e)}^{\text{grp}} + q_{t,e|g(e)}^{\text{intra}}.$$

This first selects a broad specialization regime and then refines expert preference inside that regime, which is a natural match for long-horizon behavioral evidence.

Mid stage (hybrid). Mid features evolve at every position and can deviate from long-range tendency while still preserving meaningful structure. To reflect this, the mid router combines structured and direct branches:

$$\mathbf{q}_t^{\text{mid}} = \alpha\, \mathbf{q}_t^{\text{struct}} + (1-\alpha)\, \mathbf{q}_t^{\text{direct}},$$

where $\alpha \in [0,1]$ balances regularity and flexibility. The structured branch retains group-aware prior, while the direct branch allows fast correction when current session dynamics diverge from that prior.

Micro stage (direct). Micro features are highly local and closest to the prediction target. At this scope, forcing an additional group bottleneck is often unnecessary, so direct routing is used:

$$\mathbf{q}_t^{\text{micro}} = \mathbf{q}_t^{\text{direct}}.$$

Routing weights follow the softmax rule in Eq. (2). Overall, RouteRec forms a coarse-to-fine routing pipeline in which routing structure is explicitly synchronized with temporal scope.

3.4 Training Objective

RouteRec is trained with a standard next-item objective and two router-side regularizers: route consistency and z-loss. The first regularizer shapes routing semantics, and the second stabilizes routing numerics.

Let $\mathbf{z}$ be final logits and y be the target item. The main loss is

$$\mathcal{L}_{\mathrm{CE}} = -\log \frac{\exp z_y}{\sum_j \exp z_j}.$$

For route consistency, we first compute a sample-level routing distribution at each stage by averaging token-level routing weights:

$$\bar{\boldsymbol{\pi}}_i^{\,s} = \frac{1}{L_i} \sum_{t=1}^{L_i} \boldsymbol{\pi}_{i,t}^{\,s}$$

where L_i is the valid sequence length. For each stage s, we construct a pair set $\mathcal{P}_s$ from feature-similar samples in the mini-batch and minimize

$$\mathcal{L}_{\mathrm{cons}} = \frac{1}{|\mathcal{T}|} \sum_{s \in \mathcal{T}} \frac{1}{|\mathcal{P}_s|} \sum_{(i,j) \in \mathcal{P}_s} \mathrm{JSD}\big(\bar{\boldsymbol{\pi}}_i^{\,s}, \bar{\boldsymbol{\pi}}_j^{\,s}\big)\,,$$

which encourages neighboring behavioral states to produce compatible routing distributions. In other words, semantically similar inputs should not induce arbitrarily different routing behavior.

To stabilize gating numerics, we adopt z-loss on router logits [25]:

$$\mathcal{L}_z = \frac{1}{|\mathcal{T}|} \sum_{s \in \mathcal{T}} \frac{1}{N_s} \sum_{(i,t) \in \mathcal{V}_s} \left(\log \sum_{e=1}^{E} \exp q_{i,t,e}^{\,s} \right)^2,$$

where $\mathcal{V}_s$ is the set of valid positions and $N_s = |\mathcal{V}_s|$. This term discourages excessive logit scale and helps prevent unstable, overly sharp routing. The final objective is

$$\mathcal{L} = \mathcal{L}_{\mathrm{CE}} + \lambda_{\mathrm{cons}}\, \mathcal{L}_{\mathrm{cons}} + \lambda_z\, \mathcal{L}_z,$$

where λ_{cons} and λ_z are trade-off coefficients. This objective jointly optimizes prediction quality, semantic consistency of routing, and numerical stability of expert selection.

4 Experiments

4.1 Experimental Setup

Our main model follows Section 3: macro–mid–micro stage composition, hierarchical macro routing, hybrid mid routing, direct micro routing, feature-guided learned routers, and no bias-specific routing terms. Training uses next-item cross-entropy with route consistency and z-loss, as defined in Section 3.4. Unless otherwise stated, all reported RouteRec results use this unified configuration.

We compare RouteRec against representative recurrent, graph-based, Transformer-based, frequency-aware, Mamba-based, and MoE-style sequential recommenders. The primary baseline set includes GRU4Rec, SASRec, TiSASRec, and DuoRec [5, 7, 10, 15], plus BSARec, SIGMA, DIF-SR, and FAME [11, 13, 17, 22]. When budget and space allow, we additionally include Caser, SR-GNN, BERT4Rec, FMLP-Rec, and FEARec [3, 18, 20, 21, 24] to broaden architectural coverage.

For fairness, we prioritize a unified training framework: matched embedding dimension, hidden dimension, optimizer family, batch regime, and early-stopping protocol wherever implementation permits. Data preprocessing is documented with sessionization policy, filtering thresholds, sequence truncation rules, and feature availability (timestamp/category/popularity). Evaluation follows standard top-K ranking metrics, including HR@K, NDCG@K, and MRR@K. We report MRR@20 as the primary summary metric and present other metrics for completeness.

4.2 Main Results

This subsection provides the overall performance comparison between RouteRec and all baselines across target datasets. The main table reports ranking metrics at common cutoffs and highlights RouteRec gains over strong Transformer and MoE references.

We focus analysis on three questions. First, does feature-guided routing improve over a strong shared-backbone baseline such as SASRec under matched training conditions? Second, are gains consistent across datasets with different temporal density and behavioral regularity? Third, does RouteRec remain competitive when compared with recent feature-aware or MoE-style alternatives rather than only classical baselines?

Beyond absolute ranking numbers, we summarize relative improvements against SASRec and the strongest non-RouteRec competitor on each dataset. This framing clarifies whether improvements are robust or concentrated in a narrow subset of settings.

4.3 Ablation and Diagnostic Analysis

This subsection analyzes why RouteRec works through controlled ablations and routing diagnostics. We first evaluate router-design variants (hierarchical, hybrid, direct combinations) to test whether stage-specific routing bias is necessary. We then evaluate feature-family ablations to measure the contribution of Tempo, Focus, Memory, and Exposure cues.

Next, we examine stage composition by removing or altering macro, mid, and micro branches, verifying whether coarse-to-fine routing is materially better than a single-stage or uniformly structured alternative. We also include corrupted-feature sanity checks to confirm that gains are tied to meaningful behavioral cues rather than incidental regularization effects.

Finally, we present routing diagnostics, including expert usage distribution, stage-wise route entropy, and consistency behavior under similar feature states. These analyses connect quantitative improvements to the intended mechanism: behavior-aware specialization rather than opaque capacity inflation.

5 Conclusion

The conclusion will restate the central claim of the paper: compact behavioral cues can serve as explicit routing guidance for mixture-of-experts sequential recommendation. It will summarize the empirical findings and discuss portability, interpretability, and future extensions of the framework.

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